Math 131A: Analysis
Discussion 2: Induction, Functions, and Algebraic Numbers

1 Functions

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1. Determine whether the following functions are injective or surjective.
   (i) $f : \mathbb{Z} \to \mathbb{Z}, f(n) = n + 1$
   (ii) $f : \mathbb{Z} \to \mathbb{Z}, f(n) = n^2 - 1$
   (iii) The length function $\ell : \{\text{binary strings of finite length}\} \to \mathbb{Z}_{\geq 0}$
2. A function $g : Y \to X$ is a *left inverse* to $f : X \to Y$ if $g \circ f = Id_X$.
   If f has a left inverse, show that f is injective. Is it necessarily surjective?
3. A function $g : Y \to X$ is a *right inverse* to $f : X \to Y$ if $f \circ g = Id_Y$.
   If f has a left inverse, show that f is surjective. Is it necessarily injective?

2 Proof by Induction

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1. Prove $\forall r \neq 1$,
$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$
2. Prove $\forall x \in \mathbb{R}_{>0}$ and $\forall n \geq 2$,
$$(1 + x)^n > 1 + nx$$
3. Show $\forall k, m \in \mathbb{N}$, $m \neq m + k$ from the Peano axioms.
4. **Challenge:** Consider n lines in the plane $\mathbb{R}^2$ such that no two lines are parallel and no three lines intersect at a point. Show that the number of regions that the n lines divide the plane into is $R(n) = \frac{n(n+1)+2}{2}$.

3 Algebraic Numbers & Counting

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1. Show that $\sqrt{2 + \sqrt[3]{5}} \notin \mathbb{Q}$.
2. Show that $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$
3. Prove the binomial theorem:
$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$